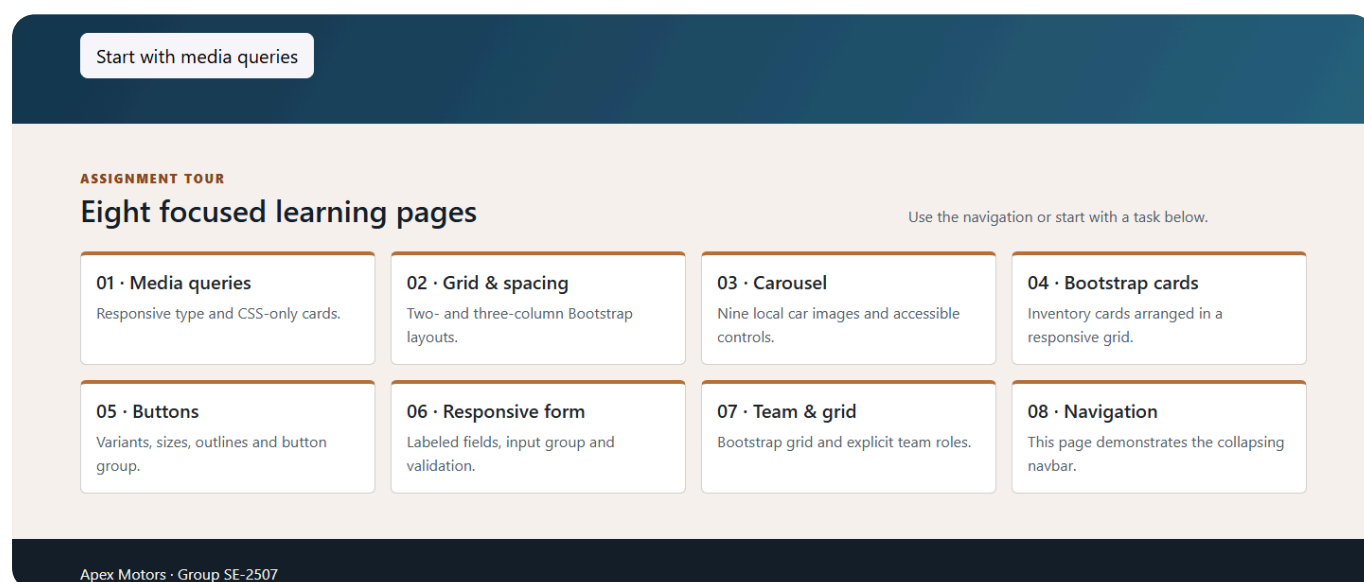


Media Queries + Bootstrap Grid

A responsive, eight-page learning site from Apex Motors. This report documents the implementation, source-code evidence, page results and team contributions.



Team: Apex Motors

Group: SE-2507

Members: Arsen · Kuttibay · Farabi · Muhammad

Course: Web Technologies / Front-End Development

Build once, adapt to every screen

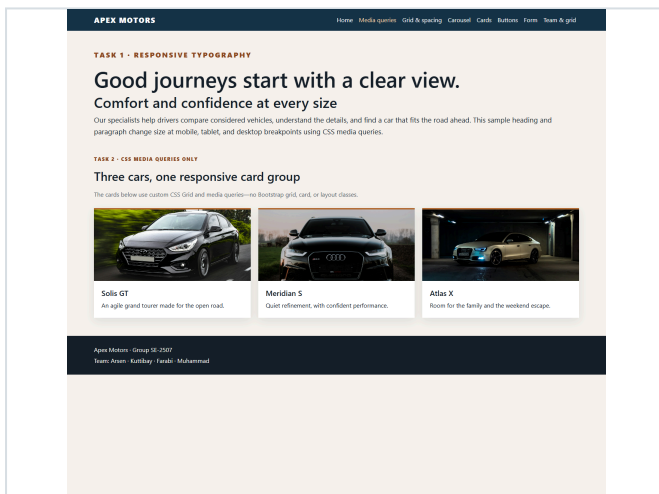
The goal was to practice responsive web design with CSS media queries and Bootstrap's 12-column grid, then apply Bootstrap components and spacing utilities to a cohesive dealership mini-site. A separate project copy keeps the existing Assignment 1/2 website unchanged.

Step-by-step implementation

1. Created the isolated assignment3-bootstrap/ project and linked eight task pages through a responsive shared navigation.
2. Added mobile, tablet and desktop media-query breakpoints for typography and a three-card layout built with plain CSS.
3. Replaced demonstration-section layouts with Bootstrap containers, rows and responsive columns; used Bootstrap spacing classes for gutters and padding.
4. Added a collapsible navbar, grouped buttons, a nine-slide local-image carousel, responsive Bootstrap cards and a labeled, validated inquiry form.
5. Reviewed semantic structure, image alternatives, contrast, visible keyboard focus and accessible names for interactive controls.
6. Captured real browser views and source excerpts, then prepared this PDF and the LMS ZIP.

Task 1 · Responsive typography

On typography-cards.html, the heading and paragraph use distinct mobile (<576px), tablet (576–991px) and desktop (≥992px) font sizes. Explicit media queries make the size changes easy to inspect and explain.



Resulting page · media queries and card group



Source excerpt · styles.css

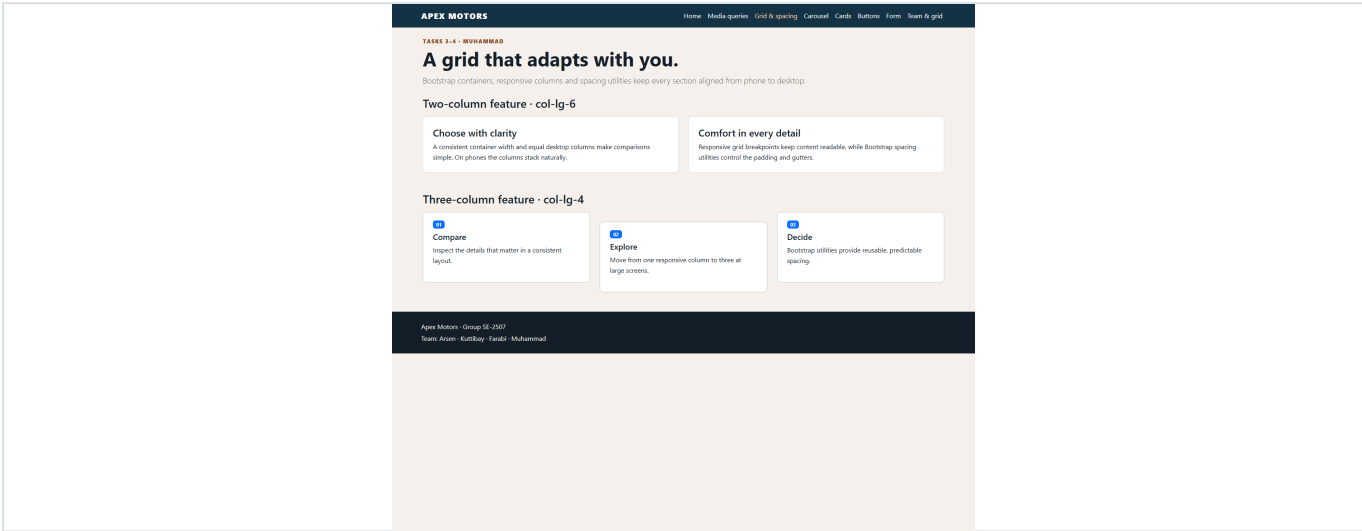
Task 2 · CSS-only card group

The three sample vehicle cards use a dedicated CSS grid: three columns on desktop, two on tablet, and one stacked column on mobile. The card group uses no Bootstrap card or grid classes; its layout is controlled by custom CSS breakpoints.

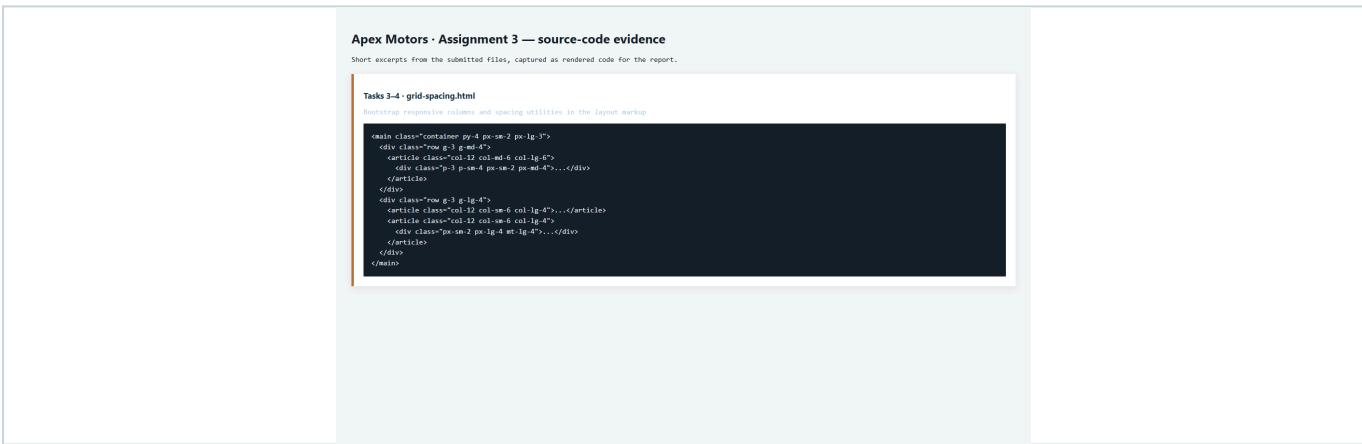
Tasks 3–4 · Grid and spacing utilities

grid-spacing.html demonstrates a two-column section using col-lg-6 and a three-column section using col-lg-4. Both start full-width on narrow phones, then use tablet and desktop breakpoints. Bootstrap container, row, and gutter classes align content consistently.

Spacing is expressed with Bootstrap utilities rather than custom layout margins and padding: examples include py-4, px-sm-2, g-md-4, and mt-lg-4. The utilities adjust at responsive breakpoints.



Resulting page · Bootstrap columns and spacing



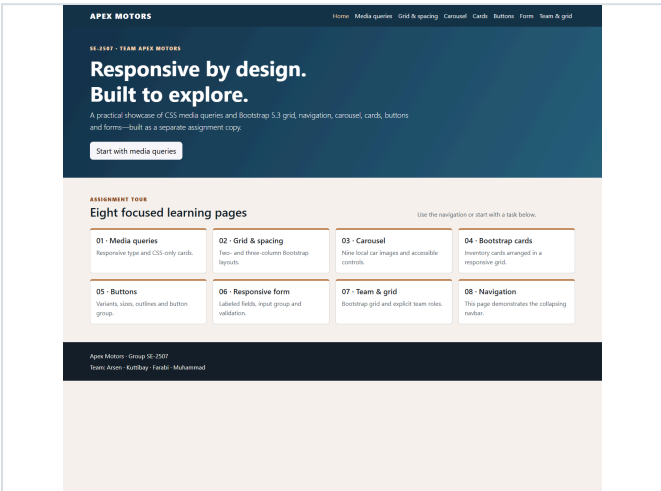
Source excerpt · responsive classes and spacing utilities

Tasks 5–6 · Navbar and buttons

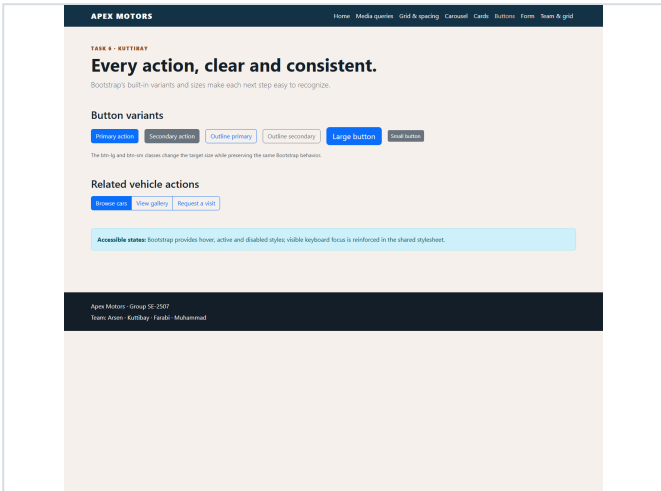
The navbar links every page, keeps the active page identifiable, and exposes its expanded/collapsed state to assistive technology. The toggler has an accessible label and target relationship.

Task 6 · Bootstrap buttons

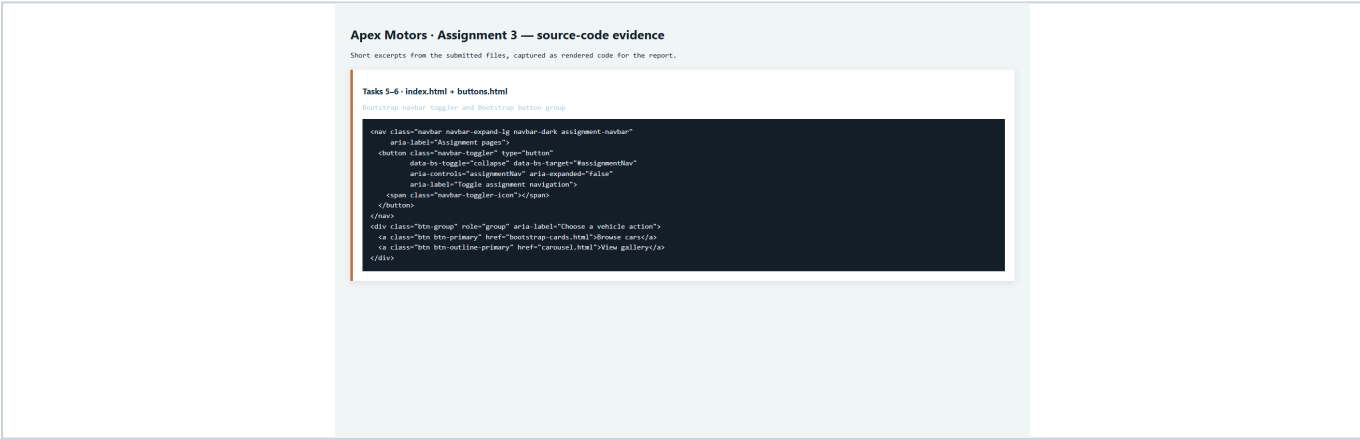
buttons.html uses Bootstrap's .btn-primary, .btn-secondary, outline variants, .btn-lg, .btn-sm, and .btn-group. Actions are links to relevant site pages or clearly identified demonstration buttons; no custom button replacement styling is used.



Resulting page · navbar and linked task overview



Resulting page · variants, sizes and related actions



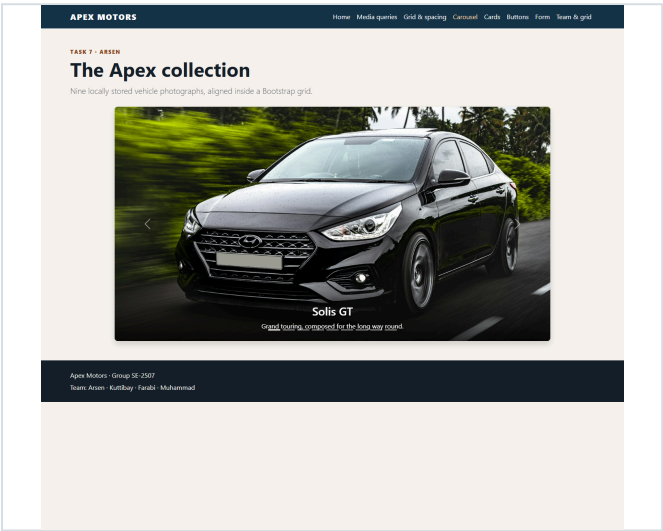
Source excerpt · Bootstrap navbar and button group

Tasks 7–8 · Explore the collection

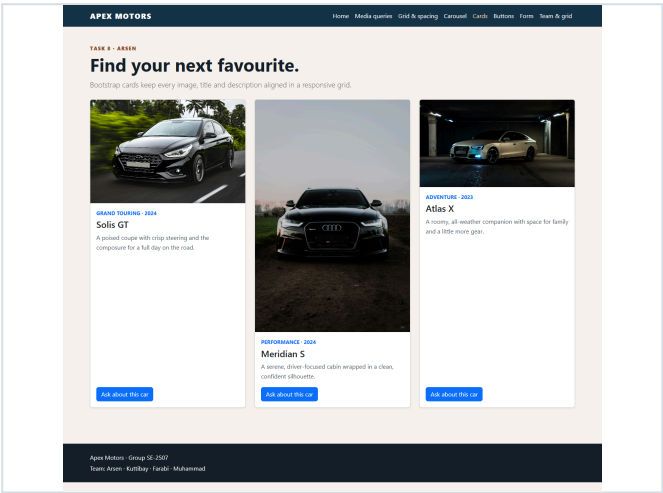
The Bootstrap carousel contains nine distinct, local vehicle images, nine matching indicators, descriptive alternative text and captions, and previous/next controls. It sits in a centered Bootstrap grid column. Motion is user-controlled rather than auto-advancing.

Task 8 · Bootstrap cards

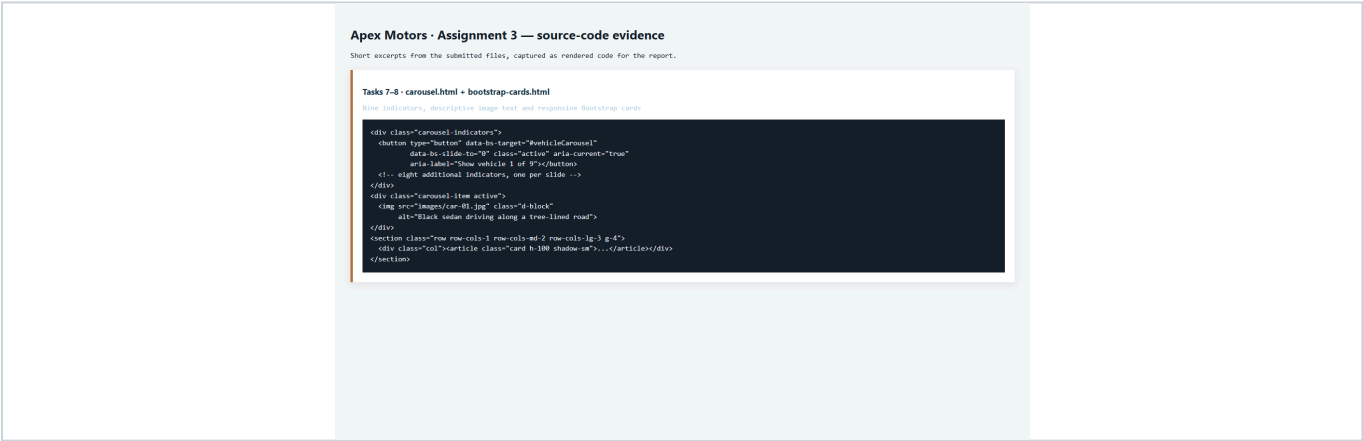
The inventory card page has three Bootstrap cards, each with an image, title, description and action. Responsive row/column classes stack on phones, make two columns on medium screens and three on large screens.



Resulting page · carousel first slide, indicators and controls



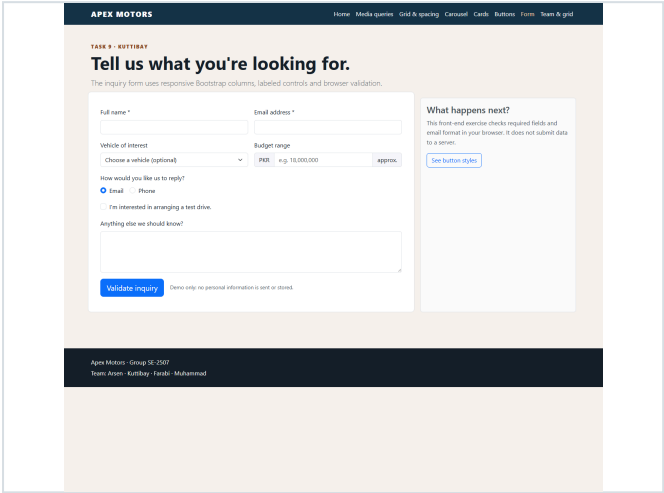
Resulting page · responsive vehicle cards



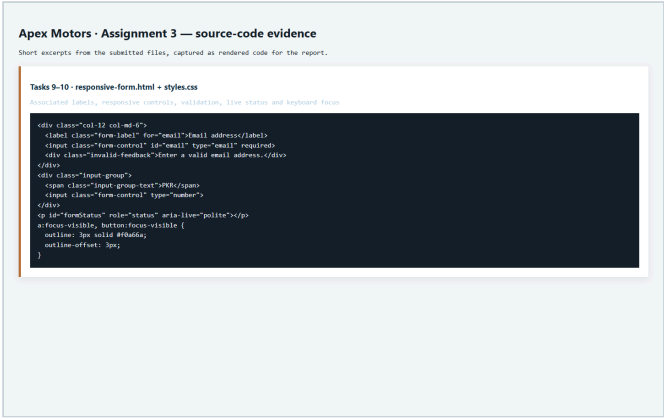
Source excerpt · carousel indicators and card grid

Tasks 9–10 · Responsive and inclusive

The inquiry page uses responsive Bootstrap columns and labeled controls, including `.form-control`, an `.input-group`, radio choices and a checkbox. Required name/email fields use native validity checks with Bootstrap invalid-feedback states. The demonstration explicitly does not transmit or store entered data.



Resulting page · labeled fields



Source excerpt · form and focus styles

Accessibility checklist

- Landmark elements, heading hierarchy, a skip link, and descriptive navigation names.
- Associated form labels, a fieldset/legend for radio choices, status feedback, and visible keyboard focus.
- Descriptive local image alternatives; sufficient contrast on dark navigation/footer surfaces; reduced-motion styling.
- Carousel indicators and previous/next controls use descriptive accessible names; carousel does not move without user action.

Team contribution map

Member	Page 1	Page 2	Bootstrap contribution
Muhammad	Typography + CSS card media queries	Grid + spacing utilities	Responsive Bootstrap grid and utility classes
Arsen	Nine-image carousel	Responsive inventory cards	Carousel and card components
Kuttibay	Buttons and button group	Responsive inquiry form	Button variants/group and form components
Farabi	Shared responsive navbar / home page	Team grid and contribution cards	Collapsible navbar, grid and cards

Live website

<https://vertex-blip.github.io/Car-dealership-website/assignment3-bootstrap/>

Final reflection

This assignment showed how CSS media queries and Bootstrap can complement each other: explicit breakpoints are useful for a small custom card group and typography exercise, while Bootstrap grid, components and spacing utilities speed up consistent layouts across a multi-page site. Local images make the carousel dependable, and accessible labels, focus styles and user-controlled movement improve the experience beyond visual responsiveness.

Each teammate can explain a pair of pages and the decisions in their component. The form remains a front-end validation example only, so a production version would need a secure, consent-aware server endpoint. The published copy is isolated from the original Assignment 1/2 pages.

Deliverables

- Eight linked HTML pages and shared CSS under assignment3-bootstrap/.
- This PDF report with screenshots of source excerpts and rendered pages.
- A ZIP archive of the complete assignment project for LMS upload.